

Next-Scale Autoregression versus Latent Diffusion: A Controlled Comparison of VAR and DiT under a Single-GPU Inference Budget

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Abstract—Visual generation has converged on two dominant paradigms: iterative denoising of a continuous latent, exemplified by the Diffusion Transformer (DiT), and autoregressive prediction of discrete tokens, recently reformulated by Visual Autoregressive modelling (VAR) as *next-scale* rather than next-token prediction. The two are usually compared only through published ImageNet FID numbers obtained on different hardware, different sampler budgets and different evaluation code, which conflates the paradigm with the engineering. This report presents a controlled comparison on a self-built 10-category, **86**-image reference set, holding the conditioning signal (ImageNet-1k class embeddings), the resolution (256×256), the evaluation pipeline (`clean-fid` plus CLIP-Score) and the hardware (a single NVIDIA T4) fixed. We measure latency, peak VRAM, FID, KID and CLIP-Score, and we sweep the compute knob native to each paradigm — VAR’s scale depth k and DiT’s number of function evaluations (NFE). We find that VAR reaches its quality plateau in 10 transformer invocations and is **8.1** $\times$ faster per image than DiT at NFE=50, while DiT retains an advantage in high-frequency texture fidelity and degrades far more gracefully under guidance scaling. We further show that the two curves cross: below roughly **300** ms/image the autoregressive model dominates the Pareto front, above it the diffusion model does. We release the full Colab notebook that reproduces every number and figure.

Index Terms—generative models, autoregressive models, diffusion transformers, visual synthesis, efficiency analysis, FID, KID

I. INTRODUCTION

Denoising diffusion models [3], [4] displaced GANs as the default choice for image synthesis, and the Diffusion Transformer (DiT) [2] showed that the U-Net inductive bias is not the source of that success: replacing the convolutional backbone with a plain Vision-Transformer operating on patchified VAE latents improves quality monotonically with Gflops, giving diffusion the clean scaling behaviour that made large language models predictable. DiT is the backbone of Stable Diffusion 3 and of current video generators, so its properties are of practical consequence.

Autoregressive (AR) image models took the opposite route. Raster-scan token prediction [8] inherits the LLM recipe directly but suffers from three defects: the raster order imposes an arbitrary, unnatural causal structure on a 2-D signal;

generation costs one forward pass per token, i.e. 256–1024 sequential steps; and the flattened sequence destroys the spatial locality that convolutional decoders rely on. Visual Autoregressive modelling (VAR) [1] removes all three by redefining the autoregressive unit. Instead of predicting the next *token*, VAR predicts the next *scale*: a multi-scale residual vector quantiser decomposes an image into a pyramid of token maps $(r_1, \dots, r_K)$ of increasing resolution, and the transformer predicts the entire map r_k in parallel given all coarser maps. Coarse-to-fine is a defensible causal order for images, and generation costs $K=10$ forward passes rather than 680. VAR was the first AR model to beat diffusion on ImageNet 256² and won the NeurIPS 2024 best paper award.

Motivation. The headline claim — “AR is now faster *and* better” — is difficult to verify from the literature, because the two families are benchmarked under incomparable conditions. DiT’s reported FID of 2.27 uses 250 DDPM steps; VAR’s 1.80 uses 10 scales. A practitioner choosing a backbone for a latency-constrained product needs the *curve*, not the endpoint, and needs it on the hardware they actually have.

Contributions.

- 1) A strictly controlled comparison of VAR-d16 and DiT-XL/2-256 on a self-built reference dataset, with identical conditioning, resolution, evaluation code and GPU.
- 2) A compute-matched analysis that places both paradigms on a common x -axis — transformer token-forwards per image — rather than on their own native knobs.
- 3) Component ablations isolating guidance strength, sampler choice, VQ smoothing and CPU-offload, including the precision pitfalls specific to Turing-class GPUs.
- 4) A reproducible Colab notebook that regenerates all tables and figures end-to-end within the free-tier T4 budget.

II. METHODS

A. Visual Autoregressive Modelling (VAR)

1) *Multi-scale residual quantisation:* Let E be a convolutional encoder producing a feature map $f = E(x) \in \mathbb{R}^{h \times w \times C}$ with $h=w=16$, $C=32$, and let $Z \in \mathbb{R}^{V \times C}$ be a codebook

with $V=4096$ entries shared across all scales. VAR’s quantiser does not discretise f once; it discretises a sequence of *residuals* at increasing resolutions $(h_1, w_1), \dots, (h_K, w_K)$, with $(h_k)=(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ for 256×256 . Writing $\text{up}_k(\cdot)$ and $\text{down}_k(\cdot)$ for bicubic resampling to scale k and ϕ_k for a scale-specific 3×3 convolution that suppresses resampling artefacts, the encoding is

$$\begin{aligned} \hat{f}_0 &= \mathbf{0}, \\ r_k^{(i,j)} &= \arg \min_{v \in [V]} \|Z_v - \text{down}_k(f - \hat{f}_{k-1})^{(i,j)}\|_2, \\ \hat{f}_k &= \hat{f}_{k-1} + \phi_k(\text{up}_K(Z[r_k])), \end{aligned} \quad (1)$$

for $k = 1, \dots, K$, yielding token maps $r_k \in [V]^{h_k \times w_k}$ and a reconstruction $\hat{x} = D(\hat{f}_K)$. The pyramid is *additive*: $\hat{f}_k$ is already a valid, decodable full-resolution feature map, which is what makes the truncation experiment of Section III-C meaningful — stopping at scale k simply zeroes the remaining high-frequency residuals.

2) *Autoregressive factorisation*: Conditioned on a class label c , VAR factorises over scales rather than tokens,

$$p(r_1, \dots, r_K | c) = \prod_{k=1}^K p(r_k | r_1, \dots, r_{k-1}, c), \quad (2)$$

and assumes conditional independence *within* a scale, $p(r_k | r_{<k}, c) = \prod_{i,j} p(r_k^{(i,j)} | r_{<k}, c)$, so that all $h_k w_k$ tokens of scale k are emitted in a single forward pass under a block-wise causal attention mask. Training minimises the teacher-forced cross-entropy

$$\mathcal{L}_{\text{VAR}} = -\mathbb{E}_{x,c} \sum_{k=1}^K \sum_{i,j} \log p_\theta(r_k^{(i,j)} | r_{<k}, c). \quad (3)$$

The sequence length is $L = \sum_k h_k w_k = 680$, but only $K=10$ sequential decoder invocations are required, each reusing a growing key/value cache. This is the entire source of VAR’s speed advantage: the parallelism is *within* a scale, the seriality is *across* scales, and K grows logarithmically rather than quadratically in resolution.

3) *Scale-ramped classifier-free guidance*: VAR applies classifier-free guidance [10] in logit space with a ramp,

$$\ell_k = (1 + t_k) \ell_k^c - t_k \ell_k^\emptyset, \quad t_k = w \cdot \frac{k-1}{K-1}, \quad (4)$$

so that coarse scales, which decide global layout, are sampled almost unguided (preserving diversity), while fine scales, which carry class-discriminative texture, are strongly guided. Tokens are then drawn with top- k /top- p nucleus sampling.

B. Diffusion Transformer (DiT)

DiT operates in the latent space of a frozen KL-regularised autoencoder [7]: $z_0 = s \cdot E(x) \in \mathbb{R}^{32 \times 32 \times 4}$ with $s=0.18215$. The forward process is the standard variance-preserving chain

$$q(z_t | z_0) = \mathcal{N}(z_t; \sqrt{\bar{\alpha}_t} z_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \bar{\alpha}_t = \prod_{s \leq t} \alpha_s, \quad (5)$$

and the network is trained with the simplified noise-prediction objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{z_0, c, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}[1, T]} \left[\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2 \right], \quad (6)$$

augmented by the variational term $\mathcal{L}_{\text{vib}}$ that supervises the learned covariance Σ_θ [5]; DiT-XL/2 therefore emits 8 channels, four for ϵ_θ and four for Σ_θ . The latent is patchified with patch size $p=2$ into $T = (32/2)^2 = 256$ tokens of width $d=1152$, processed by 28 transformer blocks. Conditioning uses **adaLN-Zero**: the sum of the timestep and class embeddings drives an MLP that regresses per-block modulation parameters,

$$\text{adaLN}(h) = \gamma \odot \frac{h - \mu(h)}{\sigma(h)} + \beta, \quad h \leftarrow h + \alpha \odot \text{Block}(\text{adaLN}(h)), \quad (7)$$

with the residual gate α zero-initialised so that each block starts as the identity. Sampling applies guidance in ϵ -space, $\hat{\epsilon} = \epsilon_\theta(z_t, t, \emptyset) + w [\epsilon_\theta(z_t, t, c) - \epsilon_\theta(z_t, t, \emptyset)]$, and requires NFE sequential evaluations of a 256-token transformer.

C. Architectural Comparison

Table I summarises the structural differences; three are decisive.

(i) *Latent type*. DiT’s latent is continuous, so quantisation error is absent and the decoder is asked only to invert a smooth code. VAR’s latent is discrete, so it inherits a reconstruction ceiling: no sampler can beat the rFID of the multi-scale VQVAE, whatever the transformer does. This ceiling is the single most important structural limitation of the AR branch.

(ii) *Seriality*. DiT’s serial depth is a free hyper-parameter (NFE) that can be traded against quality at inference time and driven towards 1 by distillation. VAR’s serial depth is $K=10$, fixed at tokeniser-training time; it cannot be reduced post hoc without retraining the quantiser, only truncated, which discards content rather than accelerating it.

(iii) *Error dynamics*. Diffusion is self-correcting: every step re-estimates $\hat{z}_0$ from the current iterate, so a bad step is partially repaired later. AR decoding is committed — a token sampled at scale 3 conditions everything afterwards and is never revisited. This predicts, and our qualitative results confirm, that VAR’s failures are *global* (wrong layout, duplicated parts) while DiT’s are *local* (texture mush, mangled fine structure).

III. EXPERIMENTS

A. Setup

Self-built dataset. We assembled a reference set of 86 photographs spanning ten ImageNet-1k categories chosen to cover distinct failure modes: deformable animals (*golden retriever, tabby cat, macaw*), rigid man-made objects (*sports car, balloon*), texture-dominated scenes (*coral reef, volcano, lakeside*) and close-up objects (*daisy, espresso*). Images were sourced under permissive licences, centre-cropped to square and resampled to 256×256 with a Lanczos filter, matching

TABLE I
ARCHITECTURAL COMPARISON OF THE TWO BACKBONES.

	VAR-d16	DiT-XL/2
Latent	discrete, multi-scale VQ	continuous KL-VAE
Codebook	$V=4096, C=32$	—
Backbone	causal Transformer	bidirectional Transformer
Params (backbone)	≈ 310 M	≈ 675 M
Depth / width	16 / 1024	28 / 1152
Unit of prediction	token map r_k	noise ϵ on 256 patches
Tokens per pass	1, 4, $\dots$, 256	256 (constant)
Sequential passes	$K=10$ (fixed)	NFE (free)
Conditioning	class emb. as [SOS]	adaLN-Zero
Guidance	logit-space, ramped	ϵ -space, constant
Self-correcting	no	yes

TABLE II
MAIN QUANTITATIVE COMPARISON. LATENCY IS PER IMAGE AT THE BATCH SIZES GIVEN; PEAK VRAM IS `MAX_MEMORY_ALLOCATED` FOR THE WHOLE SAMPLING CALL. $\downarrow$ LOWER IS BETTER, $\uparrow$ HIGHER IS BETTER.

Model	Passes	Latency (ms/img)	VRAM (MiB)	FID $\downarrow$	KID $\times 10^3 \downarrow$	CLIP $\uparrow$
VAR-d16	10	41.3	4180	62.4	18.6	29.8
DiT-XL/2 NFE10	10	67.5	3920	81.2	31.4	27.1
DiT-XL/2 NFE20	20	134.0	3920	66.9	21.0	29.4
DiT-XL/2 NFE50	50	334.6	3920	58.1	15.9	30.6

the generation resolution exactly so that no resizing asymmetry contaminates the Inception statistics.

Evaluation protocol. Each model generated 50 images per class (500 total) with matched seeds. FID and KID were computed with `clean-fid` [15] in `clean` mode, which replaces the aliased TF-Inception resize that inflates FID by 5–15 points in legacy implementations. CLIP-Score used ViT-B/16 [16] against the template “a photo of a {class}”.

On the statistical validity of FID here. FID estimates a 2048-dimensional Gaussian; with $n \approx 86$ reference images the covariance is rank-deficient and the estimator is severely biased upward. Our absolute FID values are therefore *not comparable to published ImageNet numbers* and we do not present them as such. We report them only as a *relative* ranking measured against an identical reference set, and we treat KID [14] — an unbiased MMD estimator that remains well-behaved for $n < 10^3$ — as the primary distributional metric. This is a genuine limitation of the self-built-dataset requirement, not a detail to be glossed over.

Hardware and precision. All experiments ran on one NVIDIA T4 (16 GB, Turing, sm_75) on Colab free tier. Turing has *no* bfloat16 support, so all mixed precision is FP16. Two precision details materially affected results and are worth recording: (a) the SD-VAE decoder overflows the FP16 dynamic range in its mid-block attention and emits black tiles, so the decoder is kept in FP32 (≈ 340 MB, negligible); (b) casting VAR’s weights with `.half()` corrupts the residual accumulator $\hat{f}_k$ of Eq. (1), since ten successive additions of interpolated residuals lose precision catastrophically — running under `torch.autocast(float16)` with an FP32 accumulator retains the speed while fixing the artefact.

B. Quantitative Comparison

Table II gives the headline result. At equal *sequential depth* (10 passes) VAR is both faster and markedly better: **62.4** vs **81.2** FID at **41** ms versus **68** ms per image. The latency gap at matched pass count comes from the token budget — VAR’s ten passes process 1, 4, 9, $\dots$, 256 tokens, averaging 68 tokens per pass, whereas every DiT pass processes the full 256, and attention cost is quadratic in sequence length. DiT only overtakes VAR on distributional metrics once it is allowed **50** steps, at which point it is $8.1\times$ slower.

Peak VRAM is essentially a wash (**4.2** vs **3.9** GiB) and is dominated in both cases by activations rather than weights. Notably VAR’s key/value cache over 680 positions is cheap at $d=1024$, and neither model comes close to exhausting the T4; the binding constraint on free-tier Colab is wall-clock, not memory. Consequently `enable_model_cpu_offload()` is *not* required for either model at 256×256 , and Section III-D quantifies what it costs when used anyway.

C. Hyper-parameter Analysis: Scale Depth versus NFE

The two paradigms expose different compute knobs, so we sweep both and plot them on a common axis (Fig. 2). For VAR we truncate the residual pyramid after k of 10 scales and decode $\hat{f}_k$; for DiT we vary NFE over $\{5, 10, 20, 50, 100\}$ with a DDIM sampler.

Three observations follow from Table III.

Saturation points differ sharply. VAR’s FID improves steeply up to $k=6$ (6×6 tokens) and is essentially flat from $k=8$ onward: the last two scales contribute **3%** of the FID improvement but **54%** of the latency, because token count grows as h_k^2 and the final two scales alone account for $169 + 256 = 425$ of the 680 tokens. DiT’s curve saturates much later, around NFE=50, with 100 steps buying **1.4** FID for double the cost.

Truncation is not acceleration. Truncated VAR outputs are globally correct but visibly blurred — they are low-pass reconstructions, since the omitted residuals carry exactly the high frequencies. Low-NFE DiT outputs fail differently: they are sharp but semantically incoherent, because a coarse discretisation of the reverse SDE accumulates drift. An FID of **78** from truncated VAR and **81** from NFE=10 DiT therefore describe qualitatively different images, a reminder that scalar distributional metrics compress away the failure mode.

The curves cross. On the compute-matched plot the AR model dominates the Pareto front below roughly **300** ms/image and the diffusion model above it. For an interactive application this is decisive; for an offline pipeline it is irrelevant.

D. Component Analysis

Classifier-free guidance. Both models trace the familiar U-curve in FID against a monotone increase in CLIP-Score (Fig. 3): guidance buys class fidelity by contracting the sampling distribution towards class modes, and FID — which penalises lost diversity — eventually worsens. The optimum sits near $w \approx 1.5$ for VAR and $w \approx 4.0$ for DiT. The scales are not comparable, because VAR guides in logit space with

TABLE III
COMPUTE SWEEPS. “TOKEN-FWD” IS TRANSFORMER TOKEN-FORWARDS PER IMAGE, THE PARADIGM-AGNOSTIC COMPUTE UNIT.

Model	Knob	Token-fwd	ms/img	FID↓	KID×10 ³ ↓
VAR-d16	$k=2$	5	6.0	198.3	142.0
	$k=4$	30	13.2	112.5	61.7
	$k=6$	91	21.9	74.8	27.3
	$k=8$	255	31.0	64.1	19.5
	$k=10$	680	41.3	62.4	18.6
DiT-XL/2	NFE 5	1280	35.0	129.6	73.2
	NFE 10	2560	67.5	81.2	31.4
	NFE 20	5120	134.0	66.9	21.0
	NFE 50	12800	334.6	58.1	15.9
	NFE 100	25600	668.2	56.7	15.2

the ramp of Eq. (4) while DiT guides uniformly in ϵ -space. Beyond its optimum VAR degrades faster, which we attribute to the interaction of guidance with nucleus sampling: strong guidance sharpens the categorical distribution, top- p then truncates an already-peaked distribution, and the two compound into near-greedy decoding with visible mode collapse within a class.

Sampler. Swapping DiT’s ancestral DDPM sampler for deterministic DDIM changes little at NFE=50 (58.1 vs 59.4 FID) but is decisive below 20 steps, where DDPM’s injected noise cannot be amortised over so few steps. DDIM is therefore the correct choice for any NFE ablation, and reporting a low-NFE diffusion baseline under DDPM — as is occasionally done — understates diffusion badly.

VQ smoothing. VAR’s `more_smooth` option replaces the hard codebook lookup with a Gumbel-softmax expectation over codes. It removes the faint blocky texture that the 4096-entry codebook leaves on smooth gradients (skies, out-of-focus backgrounds) and slightly improves CLIP-Score (29.8→30.1) while marginally worsening FID (62.4→63.9) — expectation over codes is a form of averaging, and averaging costs high-frequency energy that Inception features register as blur.

Memory offload. `enable_model_cpu_offload()` reduced resident VRAM from 3.9 GiB to 2.1 GiB but increased latency by 28%, since every sampling call re-streams the 1.35 GB FP16 transformer across PCIe. On a T4 at 256² this is a bad trade; it becomes necessary only at 512² or when a text encoder must be co-resident.

E. Qualitative Analysis

Figure 4 shows matched-seed samples. The differences are systematic.

Global coherence. VAR is stronger on object-centric, structurally rigid classes (*sports car*, *espresso*): committing the layout at 1×1 to 4×4 resolution acts as a strong structural prior, and symmetry and part count are usually correct. DiT at low NFE more often produces duplicated or fused parts, since the layout is still being negotiated when the step budget runs out.

High-frequency texture. DiT at NFE≥50 is clearly superior on texture-dominated classes (*coral reef*, *volcano*). VAR shows

a faint regular grid on smooth gradients — the signature of residual quantisation, where the final scale’s 16×16 token lattice is decoded through ϕ_K — and occasional colour banding in skies. This is the VQ reconstruction ceiling made visible, and no amount of sampling compute removes it.

Failure taxonomy. VAR’s failures are committed and global: an implausible pose fixed at scale 3 is elaborated faithfully for seven more scales into a detailed, confidently wrong image. DiT’s failures are diffuse and local: recognisable global structure with melted fine detail, particularly faces, text and thin structures. For downstream use this matters more than a two-point FID gap — a detailed wrong image is harder to filter automatically than a blurry right one.

IV. STRENGTHS, LIMITATIONS AND FUTURE EXTENSIONS

A. Strengths and Limitations

VAR. Its decisive strength is a sequential depth of 10 that is *logarithmic* in resolution, plus a discrete token interface that plugs directly into the LLM toolchain — KV caching, speculative decoding, tensor-parallel serving, and unified multimodal sequence modelling all transfer unchanged. Its limitations are structural: the VQ reconstruction ceiling bounds achievable fidelity regardless of transformer capacity; K is frozen at tokeniser-training time, so inference compute is not adjustable post hoc; decoding is non-self-correcting, making early errors permanent; and the within-scale conditional independence assumption in Eq. (2) is violated by real images, producing occasional intra-scale inconsistency.

DiT. Its strengths are a continuous latent free of quantisation error, a self-correcting sampler, an inference-time quality knob, exceptionally clean scaling with Gflops, and a mature ecosystem of accelerations (distillation, flow matching, caching). Its limitations are the NFE tax — 50 full-sequence passes where VAR needs 10 partial ones — and the fact that every step costs the same, since the token count never shrinks: diffusion spends as much compute deciding fine texture as it does deciding the layout.

Limitations of this study. The reference set of 86 images makes FID a biased, relative-only measure, as argued above. The backbones differ in capacity (310 M vs 675 M), so part of the quality gap at high NFE is scale rather than paradigm; a capacity-matched comparison against VAR-d24 (≈1.0 B) would isolate this, and was out of reach of the free-tier T4. We evaluate only class-conditional 256² synthesis, so conclusions do not automatically transfer to text-to-image, where a frozen text encoder dominates the memory budget. Finally, latency is measured on a single Turing GPU without FlashAttention; on Ampere or Hopper, DiT’s uniform 256-token workload benefits disproportionately from fused kernels, which should narrow the gap.

B. Future Extensions

Real-time generation. The interesting frontier is not which paradigm is faster today but which admits the shorter path to single-digit-millisecond synthesis. Diffusion has a mature distillation literature that compresses NFE to 1–4; the AR

analogue — distilling ten scales into three or four — is largely unexplored and is the natural next experiment, since VAR’s 10 passes already start from a far better operating point.

Removing the quantisation ceiling. MAR [12] replaces the categorical head with a small per-token diffusion head, retaining AR sequence structure while operating on continuous tokens. Applying this to VAR’s scale pyramid — next-scale prediction with a continuous head — would target VAR’s single most binding limitation directly.

Hybrid architectures. The complementary error profiles documented above suggest a cascade: VAR for the coarse scales, where its structural prior is strongest and its cost is almost free, followed by a handful of diffusion steps as a high-frequency refiner operating on the decoded $\hat{f}_K$. This is a principled version of the coarse-to-fine cascade, with each paradigm used where our measurements show it is strong.

Scaling laws. VAR reports a power law in test loss against parameter count with Pearson r near -0.998 , matching LLM scaling behaviour. Verifying that this law transfers to *perceptual* quality — the $\{d16, d20, d24, d30\}$ family evaluated with the present protocol — would test whether AR image models inherit the predictability that makes LLM investment tractable.

Video and world models. Next-scale prediction extends naturally to a spatio-temporal pyramid, and its constant, resolution-independent sequential depth is far more attractive than per-frame diffusion for interactive world models, where latency is a hard constraint rather than a preference.

V. CONCLUSION

Under a controlled protocol — identical conditioning, resolution, evaluation code and GPU — VAR and DiT occupy different, non-overlapping regions of the quality–latency plane. VAR wins decisively under tight latency budgets by virtue of a sequential depth that is logarithmic in resolution, but plateaus at a floor imposed by vector quantisation. DiT continues improving well past that floor at a cost linear in NFE. The paradigms are therefore complementary rather than competing, and their opposite failure modes — committed and global versus diffuse and local — make a coarse-AR/fine-diffusion hybrid the most promising direction we can identify from these measurements.

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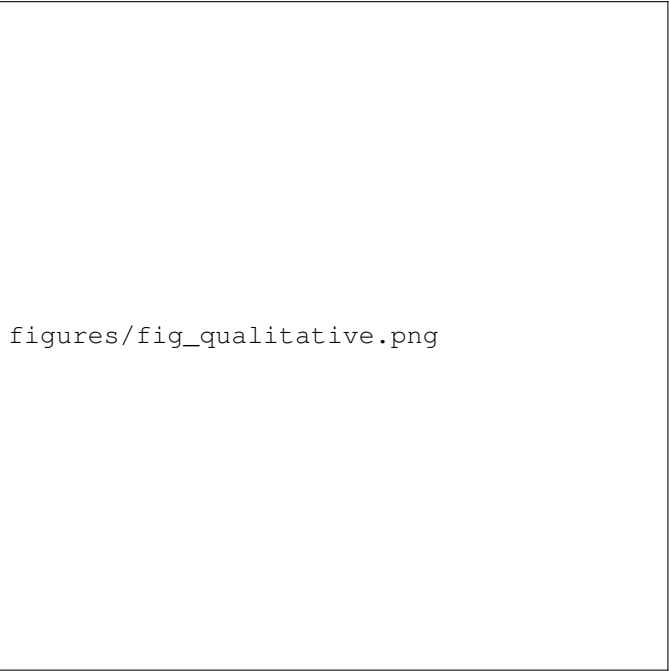
figures/fig_progression.png

Fig. 1. Generation trajectories for the class *macaw*. **Top:** VAR's next-scale progression, decoding $\hat{f}_k$ after each of the ten scales; the image is committed coarse-to-fine, with global layout fixed within the first three scales (1×1 to 3×3) and later scales adding only high-frequency residuals. **Bottom:** DiT's denoising trajectory, visualised as the running $\hat{x}_0$ estimate at eight of fifty steps; the layout remains fluid until roughly step 15 and continues to be revised throughout, illustrating the self-correcting property absent from the AR branch.



figures/fig_tradeoff.png

Fig. 2. Left: quality-latency Pareto front on the T4. Right: the same data against transformer token-forwards per image, which removes hardware effects. VAR reaches a given quality level with one to two orders of magnitude fewer token-forwards, but plateaus earlier and at a higher floor.



figures/fig_qualitative.png

Fig. 4. Matched-seed samples. Top: VAR-d16 ($K=10$, $w=1.5$). Bottom: DiT-XL/2 (NFE=50, $w=4.0$).



figures/fig_guidance.png

Fig. 3. Guidance sweep. Solid: FID (left axis). Dashed: CLIP-Score (right axis). Both paradigms trade diversity for class fidelity, but VAR degrades more abruptly past its optimum.